

THE STORY OF PRINTING

A REVOLUTION IN HUMAN COMMUNICATION CHANGES THE WORLD FOREVER

By making it possible to communicate and disseminate information at a speed and scale previously unthinkable, printing wrought changes that are still unfolding today, from triggering religious mania, to scientific and political revolutions—even changing language itself.

Printing is the impression of marks on a medium—most commonly ink on paper. The earliest writing, cuneiform, was a form of printing composed of indentations made by a stylus in clay. Printing in the modern sense of the word first arose in 8th-century China with the development of block printing.

Blocks of wood carved into bas-relief were used as stamps to reproduce multiple copies of a single text, complete with images, such as the *Diamond Sutra*, the earliest datable printed book (see 861–70).

MECHANICAL PRINTING

Block printing was laborious and slow, as each block was specific to one page. Movable type was a major advance (see panel, opposite), first achieved in

eastern Asia but perfected by German printer Johannes Gutenberg (see 1454–55). His printing press was so advanced that, except for refinements such as new typefaces and mechanization of the presses and paper handling, the basic process remained unchanged until the 19th century.

In the 1880s, the development of linotype allowed a typesetter to compose lines of type using a keyboard, rather than by hand. Stereotyping made it possible to duplicate complete pages for multiple printing. In the 20th century, filmsetting enabled rapid photographic creation of printing plates. By the end of the century, computers allowed every aspect of printing, from typesetting and graphics to inking and drying, to be done on one machine.

paper is pressed against form to produce imprint

form, or frame, for setting Gutenberg's special type

wooden coffin, which slides under platen

“ HE WHO FIRST SHORTENED THE LABOUR OF COPYISTS BY DEVICE OF MOVABLE TYPES WAS ... CREATING A WHOLE NEW DEMOCRATIC WORLD; HE HAD INVENTED THE ART OF PRINTING. ”

Thomas Carlyle, Scottish essayist and historian, 1759–1881

c. 2291–2254 BCE

Sumerian cuneiform
Stamping cuneiform inscriptions, rather than drawing them by hand, is introduced. Stamps are pressed into soft clay bricks, which are then fired.



Sumerian cuneiform

868

Block printed book
The earliest dated book (entire manuscript) is the *Diamond Sutra*, a Buddhist text found in a cave in Dunhuang, China.



The *Diamond Sutra*

1377

Movable metal type
The first metal movable type is cast in bronze in Korea and is used to produce the *Jikji Simche Yojeol*, a Buddhist scripture.



Early Korean book

8th century

Block printing
Printing using carved wooden blocks and ink is known as xylography; the earliest surviving xylographic fragment is a Buddhist *dharani* scroll from Korea.



Japanese *dharani* scroll

c. 1275–1313

Movable type
Invented in China in the 11th century, movable type is refined by Wang Shen, who uses over 60,000 wooden types in his treatise.



c. 1455–56

The printing press
Gutenberg prints the first book in Europe—the Gutenberg or 42-Line Bible (because of the number of lines on each page).

Gutenberg's printing press



1790s–1820s

Metal presses
The all-metal Columbian printing press is the first to replace the screw with levers and weights.

Columbian metal press